Prof. Eren Bilen **Name:** ______________________________

DATA 101: Final Exam

Deadline: December 18 at 5pm | Email delivery with subject line: "Name, Data 101 Final"

Instructions. Answer all questions in a jupyter notebook and turn in your assignment as a jupyter notebook file to me via email. Write clearly and show your reasoning where applicable using a combination of markdown text and code inside jupyter. You may use any material from your notes or the internet, as long as you understand and can explain all parts of the code you submit. Important: Use only the libraries we have learned in class. You will lose points if you use libraries or methods we haven't learned in class.

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1. (30 points) For this problem you will use two data sources. First is a folder that includes audio files from different genres which you can `access here`. The second source is an online repository with famous paintings from: `https://en.wikipedia.org/wiki/Wikipedia:Featured_pictures/Artwork/Paintings`.

Here is your task: Match a song from the folder provided to you with a painting from the repository above. Explain in depth why you chose this particular painting to match with the song you chose. Use arguments backed by data. Your answer should include a minimum of 250 words and 3 figures you create that best support your argument. Include all your work: text, code, figures in your answer.

2. (50 points) Read the following data set that contains newspaper articles into python and name the dataframe as `news`. The data is available `here`. Define the object `rome_keywords` as below:

```
rome_keywords = [ "rome", "roman", "empire", "imperial", "emperor", "caesar",
"augustus, ""republic", "civilization", "barbarian" ]
```

Here is your task: Create an object named `rome_counter` that is initially set to be 0. Create a for loop that goes through all articles' content in the `news` dataframe. If any of the keywords above shows up in the article, increase the `rome_counter` object by 1 for that article. For example, if an article has the word "rome" show up 3 times, and "civilization" show up 5 times, and none of the other keywords in the list, `rome_counter` should be 3+5 = 8 for that article. After your loop is done, assign `rome_counter` as a new column in your `news` dataframe and call it the same name: `rome_counter`. This new column should show how many times the keywords from `rome_keywords` has occured in the article.

Report the article with the highest `rome_counter` scores. Inspect its contents. Which keyword appeared the most from `rome_keywords` in this article? Repeat for the article with the second highest `rome_counter` score. Does the keyword distribution look the same as the first article? Are both articles talking about the Roman Empire? Why or why not?

3. (20 points) Read the dataset `books.csv` that is posted in the same folder as this exam on the course Github page and store it in a dataframe object called `books`.

(a) Who is the author whose books are included in this data set?

(b) What is the publication date of the author's first book? How old were they? What is the publication date of the author's most recent book? How old were they? Calculate the duration of the author's career so far.

(c) Make a scatterplot with age on the x axis and average rating on the y axis. Do you notice a correlation? Is it positive, negative, or unclear?

(d) Fit a linear regression with age as the only predictor variable and average rating as the outcome variable. Interpret the coefficient of age. Is it positive, or negative? Is it statistically significant? What does its magnitude suggest?

(e) Fit a quadratic regression with age and age squared as predictor variables and average rating as the outcome variable. Interpret your results. Are the coefficients for age and age squared statistically significant? What does the shape of the quadratic fit suggest about the author's ratings?